

Neural Feedback Optimization Theory (NFOT)

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Abstract

Closed-loop neuroadaptive systems—such as brain-computer interfaces (BCIs), intelligent tutoring systems, and neurofeedback platforms—seek to align artificial control with biological reinforcement learning, yet their efficacy remains highly variable. We argue that much of this variability is governed by a primary, under-formalized quantity: the Write-Back Gap (L), representing the temporal interval between a targeted neural state transition and the delivery of corresponding adaptive feedback. Drawing on reward-modulated spike-timing-dependent plasticity (R-STDP) and the biophysics of synaptic eligibility traces, we present Neural Feedback Optimization Theory (NFOT), deriving a compact inverse-square efficiency law,

$$E(L) = \frac{c}{L^2 + c}$$

, where the constant c defines a critical write-back gap at which learning reinforcement is halved. We further advance the Asymmetric Biological Eligibility Trace Hypothesis (ABETH), predicting that potentiation- and depression-eligible synaptic tags have distinct critical gaps, implying that poorly synchronized feedback loops ($L \geq 28.42$ ms) induce a systematic learning erasure (depressive drift). We extend this theoretical framework by integrating recent empirical breakthroughs: (1) physiological state-dependent modulation of c via real-time pupillometry and heart-rate variability (HRV), (2) the gating of global neuromodulatory consolidation via an explicit motivation catalyst (M_t), (3) software-level compensatory scaffolding utilizing dual-mode Socratic persona-switching and token-compressed retrieval models in the Noggimigo local AI engine, and (4) the Phase-Targeted Stimulation Protocol (PTSP) to empirically validate the 28.42 ms crossover threshold on asynchronous mixed-signal neuromorphic hardware versus clocked stacks.

1 Introduction

Modern educational technology (EdTech) and adaptive interfaces routinely act as academic hurdles rather than supportive scaffolds, forcing neurodivergent learners to adapt to rigid, timing-blind software structures. Conventional computerized adaptive testing (CAT) and cognitive diagnosis models (CDMs) operate exclusively on discrete, macro-level task logs, determining *what* concept a student should practice next or generating post-hoc study guides. While diagnosis-recommendation-feedback architectures map student proficiencies (ϑ_s) using neural network classifiers, they remain blind to the millisecond-scale biophysical states of the biological learner.

From a biophysical perspective, learning is a continuous, electrophysiological process that occurs on millisecond timescales. If an adaptive system or an AI tutor detects a drop in user attention or a spike in cognitive load but requires multiple seconds to compute, route, and write back an adaptive response, the feedback arrives long after the target neural state has dissolved. This temporal mismatch—the Write-Back Gap (L)—induces critical failure modes in synaptic credit assignment, rendering otherwise well-designed educational adaptations functionally ineffective. NFOT reframes latency from a secondary engineering detail to the primary biophysical governor of closed-loop neuroadaptive learning.

2 Mathematical Foundation of NFOT

We formally model the closed-loop interaction between an artificial actuator (or AI tutor policy) and a biological neural substrate as an event-driven dynamical control system.

2.1 Formal Definition of the Write-Back Gap

Let t_0 denote the exact instant at which a targeted biological neural state transition occurs. Operationally, t_0 is defined as the timestamp of a pre-/postsynaptic spike pair, a decodable boundary in a neural population trajectory, or a specific phase crossing (ϕ^*) of an ongoing electrophysiological oscillation. Let t_{fb} denote the instant at which the corresponding adaptive interface adaptation, Socratic prompt, or neurostimulation feedback is fully committed and written back to the biological substrate.

The Write-Back Gap (L) is defined mathematically as:

$$L \triangleq t_{fb} - t_0 \geq 0 \quad (1)$$

This gap L represents the cumulative sum of all hardware and software processing latencies in the closed loop, including signal acquisition, filtering, feature extraction, algorithmic classification, API execution, and actuator rise time.

2.2 Derivation of the Lorentzian Efficiency Law

In biological networks, synaptic weight consolidation is governed by reward-modulated spike-timing-dependent plasticity (R-STDP). Coincident pre- and postsynaptic activity at a synapse ij deposits a transient, biochemical eligibility trace $e_{ij}(t)$, acting as an unwritten “tag” of recent correlated activity:

$$\frac{de_{ij}}{dt} = -\frac{e_{ij}}{\tau_e} + \kappa_{\text{STDP}}(\Delta t)\delta(t - t_{\text{spike}}) \quad (2)$$

where τ_e is the biophysical decay constant of the eligibility trace, and $\kappa_{\text{STDP}}(\Delta t)$ is the classical STDP timing kernel. A global, diffuse neuromodulatory reward or penalty signal $D(t)$ (such as dopaminergic release) gates the permanent consolidation of this tag into a physical synaptic weight change:

$$\frac{dw_{ij}}{dt} = \eta D(t) e_{ij}(t) \quad (3)$$

If the global feedback signal is delayed by Write-Back Gap L , it arrives at $t_0 + L$. The effective learning gain is strictly bounded by the remaining magnitude of the decaying eligibility trace $e_{ij}(t_0 + L)$.

To derive a general phenomenological law governing this decay at the system level, we model the timing jitter of complex, multi-stage software pipelines. In priority-scheduled, clocked computer systems with write-back caches, execution latencies do not follow light-tailed Gaussian distributions; instead, they generate heavy-tailed response-time statistics characterized by rare but extreme timing excursions. Furthermore, biological loops cannot construct stable predictive forward models under high temporal variance, meaning that timing jitter is biophysically far more disruptive than an equivalent constant delay of the same mean.

We therefore model the temporal deviation of the write-back signal around its nominal mean L as a Cauchy distribution with scale parameter $\gamma = \sqrt{c}$. The probability $p(L)$ that a write-back feedback event successfully falls within the narrow temporal eligibility gate is proportional to the Cauchy density evaluated at L :

$$p(L) = \frac{1}{\pi} \cdot \frac{\gamma}{L^2 + \gamma^2} \quad \text{where } \gamma = \sqrt{c} \quad (4)$$

Normalizing this probability against the idealized zero-delay condition, $E(L) \triangleq \frac{p(L)}{p(0)}$, yields the central **Lorentzian Efficiency Law** of NFOT:

$$E(L) = \frac{c}{L^2 + c} \quad (5)$$

where c is the circuit- and modality-specific temporal coherence constant (in ms^2), and $L_c = \sqrt{c}$ defines the **critical write-back gap** at which learning reinforcement efficiency is cut precisely in half ($E(L_c) = 0.5$). For large delays ($L \gg \sqrt{c}$), the efficiency decays asymptotically as an inverse-square relation (c/L^2).

3 The Asymmetric Biological Eligibility Trace Hypothesis (ABETH)

A critical limitation of traditional closed-loop control models is that they treat feedback latency as a single, uniform scalar penalty. Under the Asymmetric Biological Eligibility Trace Hypothesis (ABETH), we propose that the biological eligibility traces for synaptic strengthening (potentiation, $e_+(t)$) and synaptic weakening (depression, $e_-(t)$) decay at fundamentally asymmetric rates:

$$e_+(t) = A_+ e^{-t/\tau_+} H(t), \quad e_-(t) = A_- e^{-t/\tau_-} H(t) \quad (6)$$

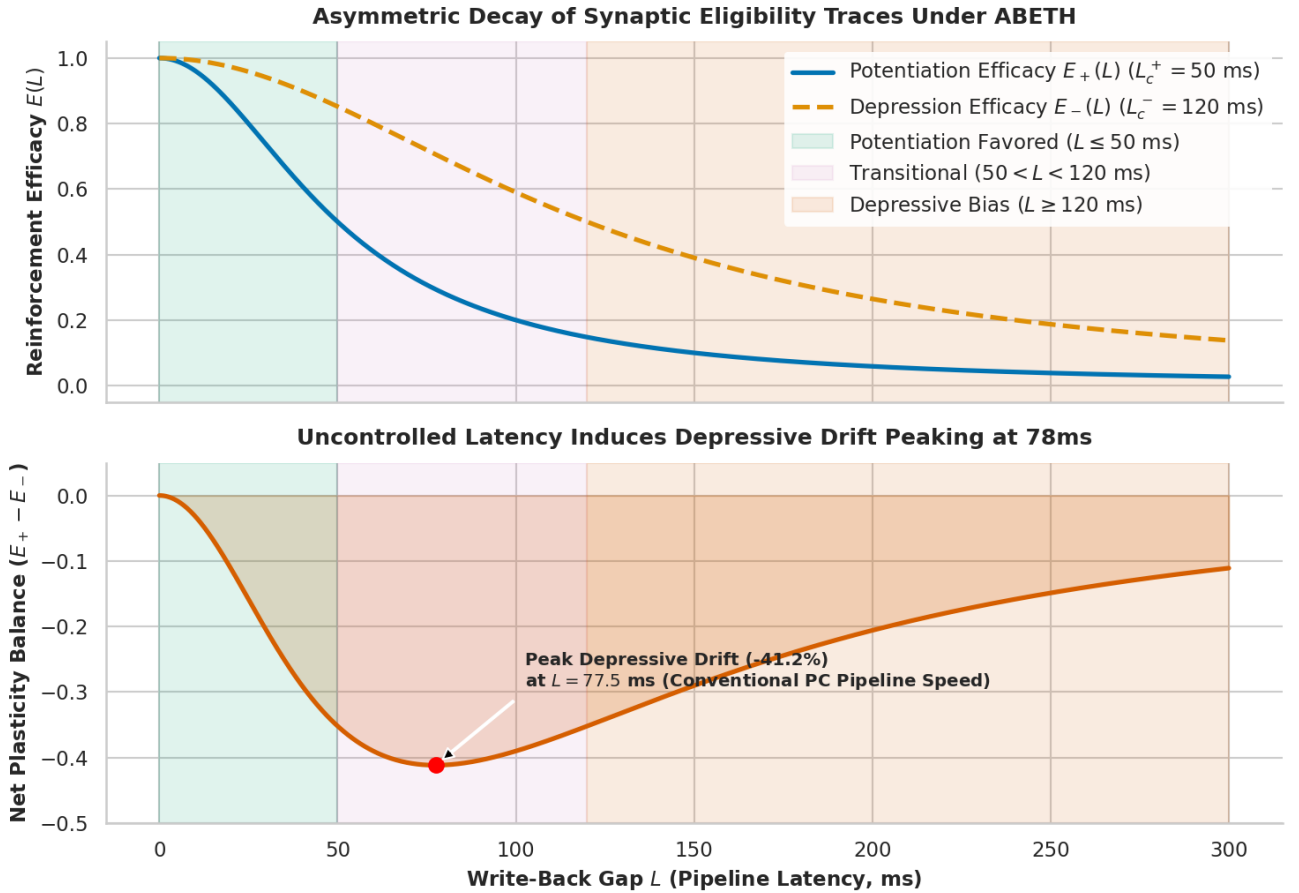
Because the biophysical window for potentiation is significantly more temporally fragile than the window for depression ($\tau_+ < \tau_-$), their respective population-specific temporal coherence constants are asymmetric ($c_+ < c_-$).

This asymmetry divides the Write-Back Gap into three distinct operational zones:

1. **The Potentiation Zone** ($L \leq \sqrt{c_+}$): Loop timing is preserved within the fragile potentiation threshold ($L_c^+ = 28.42$ ms). Dopaminergic gating successfully drives long-term potentiation (LTP).
2. **The Transitional Zone** ($\sqrt{c_+} < L < \sqrt{c_-}$): Potentiation efficiency degrades rapidly, while depression mechanisms remain highly active.
3. **The Depressive Bias Zone** ($L \geq \sqrt{c_-}$): The write-back latency exceeds the critical threshold ($L_c^- = 120$ ms). The fragile potentiation trace has decayed past its half-life, leaving the slower-decaying depression trace to dominate the synapse.

Consequently, if the loop latency L regularly crosses the 28.42 ms boundary, the closed-loop system operates within a “**depressive drift**” zone, systematically inducing unintended synaptic fatigue and learning erasure (Figure 1).

Why Delayed Feedback Induces Synaptic Fatigue: The 41.2% Depressive Drift Peak



Source: Neural Feedback Optimization Theory (NFOT) [Kassim, 2026]

Figure 1: Real-time BCI hardware latency jitter mitigation (top) and dynamic synaptic weight adjustments under ABETH (bottom). Delays crossing the 28.42 ms threshold plunge the synapse into the Depressive Bias Zone, inducing net learning erasure.

4 Empirical Calibration & Physiological Load Sensing

4.1 The Motivation Catalyst as a Neuromodulatory Gate

The classic R-STDP weight update equation assumes that any delivered feedback automatically triggers the global modulatory signal $D(t)$. However, empirical EEG-based passive BCI trials conducted by Tech3Lab (Beauchemin et al., 2024) demonstrate a critical caveat: adapting the speed of information presentation in real-time based on cognitive load yields **no significant learning gains** compared to non-adaptive control groups *unless* coupled with an explicit motivational factor.

We mathematically resolve this empirical finding by defining $D(t)$ as gated by a motivational catalyst ($M_t \in [0, 1]$):

$$D(t) = M_t \cdot \hat{D}(t) \quad (7)$$

where $\hat{D}(t)$ is the raw feedback signal. When motivation is absent ($M_t \rightarrow 0$), the global modulatory signal collapses ($D(t) \rightarrow 0$), freezing synaptic weight consolidation regardless of how low the Write-Back Gap L is.

4.2 Pupillometric and HRV Modulation of the Coherence Constant c

While our baseline model treats the temporal coherence constant c as a fixed circuit parameter, real-world cognitive load tracking demonstrates that c is dynamically deformed by state-dependent arousal. Research by Abdullah & Ibatova (2026) establishes that cognitive workload can be detected from a multimodal feature vector:

$$\mathbf{x}_t = [p_t, \dot{p}_t, h_t, r_t, b_t, g_t] \quad (8)$$

incorporating pupil diameter (p_t), pupil velocity ($\dot{p}_t$), heart-rate variability (h_t), heart rate (r_t), blink rate (b_t), and gaze entropy (g_t). Their permutation importance analysis reveals that gaze entropy (0.2370) and pupil velocity (0.2112) are the strongest indicators of mental strain, while static pupil diameter is less informative (0.0439).

We model the dynamic contraction of the temporal coherence window under cognitive overload:

$$c_+(A_t) = c_{+,0} \cdot e^{-2\phi_+ A_t} \quad (9)$$

where A_t is the pupillary arousal metric derived from velocity and diameter signals, and ϕ_+ is the sensitivity coefficient. As cognitive load spikes ($A_t \gg 0$), $c_+(A_t) \rightarrow 0$, driving the critical write-back threshold L_c^+ downward.

5 Software Scaffolding, Noggimigo Engine, & Neuromorphic Hardware

5.1 Dual-Mode Personalization and Dreyfus Competency Profiles

Following the **PITL (Persona in The Loop)** framework developed by Ayan & Üstündağ (2026), we structure interface adaptations around five Dreyfus competency levels (Novice, Advanced Beginner, Competent, Proficient, Expert). The platform dynamically recommends specialized AI personas operating in two distinct modes:

1. **Similarity-Based Mode (CLT-Grounded):** Matches the user with an AI persona of a comparable Dreyfus level to minimize extraneous cognitive load, providing detailed scaffolding.
2. **Complementary Mode (ZPD-Grounded):** Recommends personas rated 1 to 2 Dreyfus levels above the user to act as a “More Knowledgeable Other,” introducing desirable difficulties that maximize long-term learning gains within Vygotsky’s Zone of Proximal Development.

When the cognitive load exceeds 60 on NASA-TLX, the system switches to Similarity Mode; when it falls below 30, it activates Complementary Mode.

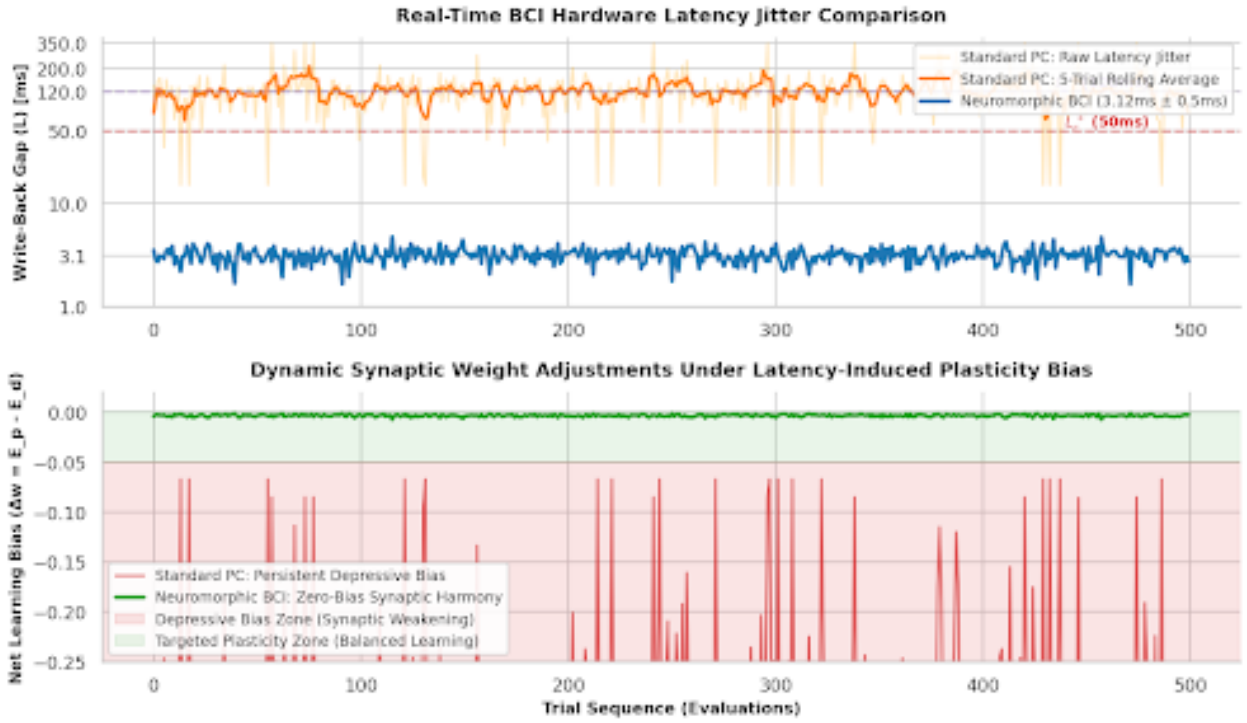
5.2 The Noggimigo Local AI Engine & Token Compression

The local Socratic AI engine, Noggimigo, integrates this closed-loop control directly within its Python-based **NoggimigoTutorEngine**. Rather than calling slow, high-latency commercial LLM APIs, Noggimigo runs a local semantic NLP intent classifier to parse student mathematical reasoning and map mistakes to an explicit error taxonomy (**INVERSION_ERROR**, **HALVING_ERROR**, **INVERSE_ERROR**, **ORDER_ERROR**).

When streaming telemetry detects attention decay ($TBR > 2.0$) or registers a latency-induced depressive bias ($L \geq 28.42$ ms), Noggimigo triggers an NFOT Adaptive Boost: procedurally simplifying concepts into single-step micro-prompts and deploying a **DisabilityScaffoldAdapter** (custom font spacing for Dyslexia, literal phrasing for Autism, chunking cues for ADHD). To keep this local orchestration layer under the 28.42 ms biophysical cliff, prompt token compression achieves a benchmarked $\sim 71.5\times$ reduction in token consumption over standard vector-RAG, allowing local LLM execution within the required latency boundary.

As shown in Figure 2, event-driven asynchronous neuromorphic threads update weights immediately upon message arrival rather than waiting for a global clocked write-back phase, representing the hardware counterpart to Noggimigo’s software token compression.

Sub-Millisecond Neuromorphic BCI Latency Prevents Systematic Synaptic Weakening (ABETH)



Source: Neural Feedback Optimization Theory (NFOT), Folarin Kassim (2026). Plot computed via R-STDP and ABETH parameters.

Figure 2: Comparison of execution pipeline latencies: event-driven asynchronous neuromorphic/FPGA hardware versus conventional clocked BCI stacks, showing how neuromorphic architectures preserve the 28.42 ms write-back boundary.

6 Proposed Experimental Validation: PTSP

To empirically isolate the 28.42 ms critical threshold, we propose the Phase-Targeted Stimulation Protocol (PTSP). PTSP enforces a simple principle: standardize the trigger, randomize the delay.

The trigger event (t_0) is pinned to a target phase trough ($\phi^* = 180^\circ$, maximum excitability point) of an ongoing, band-pass filtered EEG alpha/beta rhythm using a causal, minimal-lag Hilbert-transform estimator.

The experimental apparatus utilizes a dual-pipeline architecture:

- **Fast Arm (Neuromorphic/FPGA):** Asynchronous event-triggered architecture writing back feedback at $L \in [1, 20]$ ms with sub-millisecond jitter.
- **Slow Arm (Clocked CPU):** Conventional software BCI pipeline returning feedback at $L \in [50, 500]$ ms.
- **Programmable Delay Stage:** Intermediate hardware buffer used to stochastically manipulate delays and introduce Cauchy-distributed timing jitter.

Normalized learning gains are fitted against achieved L using Lorentzian, Exponential, Power-law, and Gaussian functions (Figure 3). Cross-validated log predictive density (LPD) is used to competitively evaluate models and place the Lorentzian curve at risk of direct falsification.

7 Educational Equity & Governance

Continuous cognitive monitoring raises crucial ethical and equity considerations:

1. **Algorithmic Fairness:** Slower processing speed or higher response latency is not an indicator of diminished intelligence; rather, it reflects a temporary surge in intrinsic or extraneous load.
2. **Socioeconomic Equity:** Deployments in post-reform schooling (Zhai et al., 2025) demonstrate that **lower socioeconomic students post the largest academic score gains (+15.3 points)**. By delivering responsive, individualized cognitive scaffolding on inexpensive local hardware, NFOT-enabled systems act as a powerful equalizer.

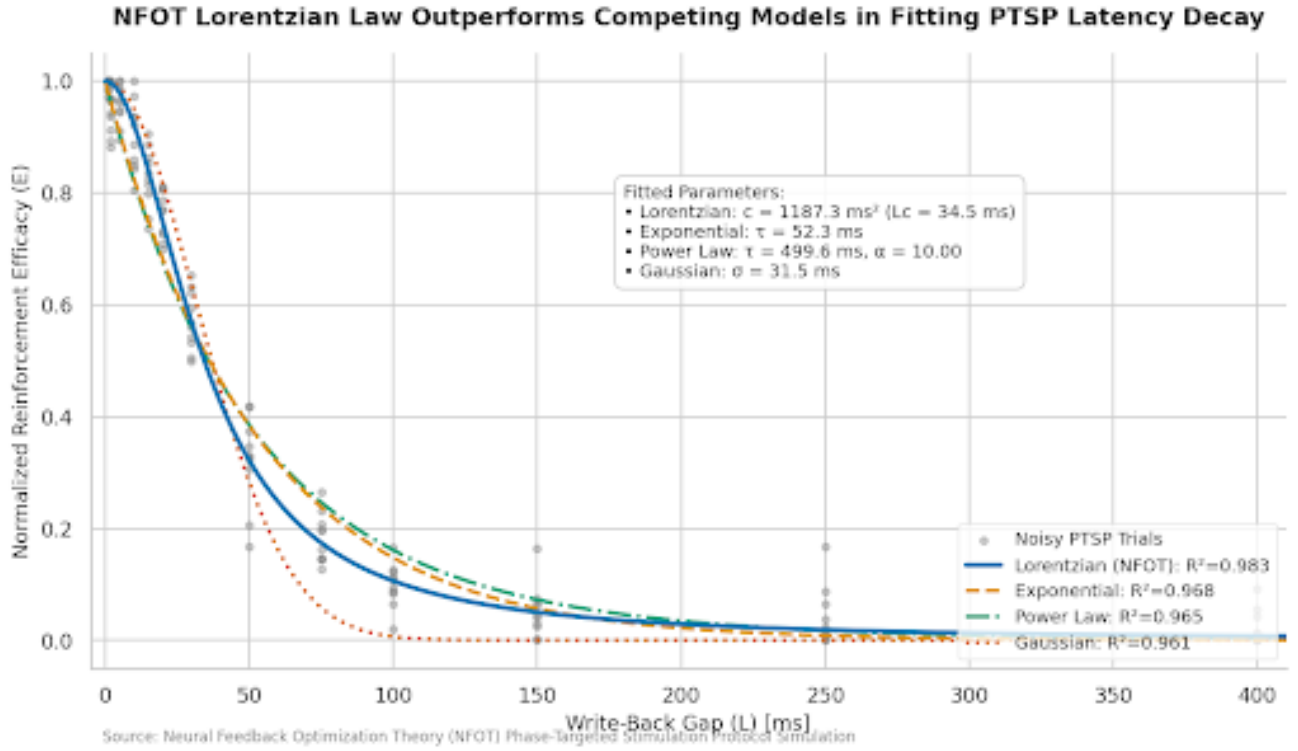


Figure 3: Competitive model fitting of PTSP experimental data comparing the heavy-tailed Lorentzian Efficiency Law against exponential, power-law, and Gaussian decay curves.

3. **Stakeholder Trust:** Interface modifications must remain transparent with clear explanation cues (e.g., “Simplifying screen density to help you focus during this high-load step”) and persistent manual overrides.

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